

Histone levels are regulated by phosphorylation and ubiquitylation dependent proteolysis

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Histone levels are tightly regulated to prevent harmful effects such as genomic instability and hypersensitivity to DNA-damaging agents due to the accumulation of these highly basic proteins when DNA replication slows down or stops. Although chromosomal histones are stable, excess (non-chromatin bound) histones are rapidly degraded in a Rad53 (radiation sensitive 53) kinase-dependent manner in *Saccharomyces cerevisiae*. Here we demonstrate that excess histones associate with Rad53 in vivo and seem to undergo modifications such as tyrosine phosphorylation and polyubiquitylation, before their proteolysis by the proteasome. We have identified the Tyr 99 residue of histone H3 as being critical for the efficient ubiquitylation and degradation of this histone. We have also identified the ubiquitin conjugating enzymes (E2) Ubc4 and Ubc5, as well as the ubiquitin ligase (E3) Tom1 (temperature dependent organization in mitotic nucleus 1), as enzymes involved in the ubiquitylation of excess histones. Regulated histone proteolysis has major implications for the maintenance of epigenetic marks on chromatin, genomic stability and the packaging of sperm DNA.